

Mikayla Newbrey

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Education

University of Massachusetts Amherst

PH.D. STUDENT, ORGANISMAL & EVOLUTIONARY BIOLOGY PROGRAM

Amherst, Massachusetts

2025 - Present

- Advisor: Dr. Lisa Komoroske
- Expected 2031

University of Massachusetts Amherst

BACHELOR OF SCIENCE IN BIOLOGY

Amherst, Massachusetts

2022-2024

Cape Cod Community College

ASSOCIATE OF ARTS IN BIOLOGY

Barnstable, Massachusetts

2019-2022

Work & Research Experience

PhD Research

ENVIRONMENTAL CONSERVATION DEPARTMENT, UNIVERSITY OF MASSACHUSETTS AMHERST

Fieldwork: Gloucester, MA

2025 - Present

- Conduct population genomics and kinship analyses using high-throughput sequencing data
- Develop SNP panels for relatedness and population structure using GT-seq approaches
- Conduct riverscape genomics analyses to identify environmental drivers of population structure and adaptive variation

Lab Manager

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2024-2025

- Directed lab operations, including protocol development, sample preparation, and equipment management
- Coordinated multiple research projects and maintained workflow efficiency across experiments
- Trained and mentored undergraduate and graduate researchers in molecular and genomic techniques

Lead Lab Technician

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2022 - 2025

- Optimized Nanopore sequencing workflows through experimental testing and performance evaluation.
- Executed large-scale genomics projects, including library preparation, sequencing, and downstream analysis.
- Developed and maintained reproducible pipelines for genomic data processing.

Lab Technician

CHENG LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

2022 - 2023

- Supported experiments on local adaptation and environmental stress responses in marine organisms.
- Maintained animal systems and conducted daily monitoring under controlled conditions.
- Assisted with common garden experiments and data collection.

CAFE Scholars Internship

MOLECULAR ECOLOGY AND CONSERVATION LAB, UNIVERSITY OF MASSACHUSETTS AMHERST

Amherst, MA

June – Aug 2023

- Conducted population genetics analyses to assess relatedness in marine turtles.
- Validated parent–offspring assignments using genomic markers.
- Performed statistical analyses to support conservation-focused research.

WHOI, CC-CREW Internship

TEPOLT LAB, WOODS HOLE OCEANOGRAPHIC INSTITUTION

Woods Hole, MA

June– Aug 2022

- Designed a 213-SNP panel to resolve population structure in invasive European green crabs.
- Implemented targeted genotyping workflows for large-scale population analysis.
- Integrated transcriptomic data to identify signatures of adaptation.

WHOI, Blue Economy Internship

TEPOLT LAB, WOODS HOLE OCEANOGRAPHIC INSTITUTION

Woods Hole, MA

Jan - June 2022

- Investigated thermal tolerance in invasive European green crabs using respirometry and genomic approaches.
- Analyzed physiological and genetic responses to environmental stress.

Scientific Publications

In Review

Dwane, C., L.M. Komoroske, A.L. Rugila, B.P. Bentley, E. Rawson, E. Clark, G. Nichols, **M. Newbrey**, E. Bucari, C. Yan, J. Barley, R. Horrigan, L. McCarthy, N. Duncan, A.B. Juarez Stucker, A.R. Villeneuve, B.S. Cheng (2025) Adaptation to warm environments with a fast pace of life in a marine predatory snail. Preprint | In revision at Ecological Monographs

In Prep

Venkataraman, Y.R., S.K. Shapiro, **M. Newbrey**, C.K. Tepolt. Metabolomic and lipidomic shifts underpin physiological acclimation to thermal stress in the European green crab (*Carcinus maenas*). Manuscript in preparation.

Teaching & Outreach

Introduction to Ecology

TEACHING ASSISTANT

UMass Amherst

Spring 2025

- Instructor: Tristram Seidler

Supplemental Instructor

CAPE COD COMMUNITY COLLEGE

Barnstable, MA

2020-2022

- Collaborated with faculty to develop course materials, including lecture notes, problem sets, and solution guides, to enhance student understanding of complex topics.
- Planned and facilitated study sessions for General Chemistry, General Biology, and Survey of Biology, reinforcing core concepts and fostering collaborative learning.

Student Tutor

CAPE COD COMMUNITY COLLEGE

Barnstable, MA

2020-2022

- Conducted peer-led review sessions to clarify lecture material, address student questions, and prepare students for exams in General Chemistry, General Biology, and Survey of Biology.
- Delivered one-on-one and group tutoring sessions, tailoring content and teaching approaches to accommodate individual learning styles and academic goals.

MENTORSHIP

Cammi Cooper

UMass Amherst

- Undergraduate, Five College Marine Science Program

2025 - Present

Annalise Kuhlmann

Mount Holyoke College

- Undergraduate, Five College Marine Science Program

Fall 2024

Mary Rutter

UMass Amherst

- Undergraduate, Animal Science Independent Study
- Post-Baccalaureate Research Technician

Jan – July 2024

Mikayla Donovan

UMass Amherst

- Undergraduate, Biology Independent Study
- Lee-SIP Summer Scholar Mentee

Jan – May 2024

Emma Ferreira

UMass Amherst

- Undergraduate, Animal Science-Biotechnology Independent Study

2023

Grants & Fellowships

TOTAL FUNDING RAISED: \$7,389

CNS Graduate Student Fellowship

Fall 2025

UMASS AMHERST - GRADUATE SCHOOL (\$7,389)

Awards

Placed (3rd) Conference-wide Poster Competition

Baltimore, Maryland

INTERNATIONAL CONFERENCE ON MARINE BIOINVASIONS

2023

Presentations

CONFERENCES

A Genetic Toolkit for Understanding the Rapid Expansion of European Green Crabs in the Northeast Pacific

Baltimore, Maryland

INTERNATIONAL CONFERENCE ON MARINE BIOINVASIONS

2023

Quantifying Relatedness of Green Sea Turtles in the Hawaiian Islands

UMass Amherst

CAFE SCHOLARS PROGRAM POSTER CONFERENCE

2023

Professional Development & Relevant Coursework

Landscape Genomics

Scheduled, Summer 2026

PHYSALIA WORKSHOP

Applied Ecological Statistics

Spring 2026

DR. MEG GRAHAM MACLEAN, UNIVERSITY OF MASSACHUSETTS AMHERST

Assembling Genomes with Oxford Nanopore: A Hands-On Guide to Long-Read Sequencing

Spring 2024

PHYSALIA WORKSHOP

Ecology of Fish

Fall 2023

DR. ANDREW DANYLCHUK, UNIVERSITY OF MASSACHUSETTS AMHERST

Skills

Software & Tools	Programming & Computing	Wet Lab Techniques	Fieldwork
QGIS	R (RStudio)	GT-Seq	Boat operation and trailering
ArcGIS Pro	Python	RAD-Seq	Davit operation
ImageJ	Bash	Whole Genome Resequencing	Seining
Zotero	SLURM	Nanopore Sequencing	Net tows

Certificates

Scientific Diver

2025 (Active)

AMERICAN ACADEMY OF UNDERWATER SCIENCES

- Certified in scientific diving, including underwater data collection, species surveys, and equipment deployment under research protocols

Rescue Diver

2025 (Active)

NATIONAL ASSOCIATION OF UNDERWATER INSTRUCTORS

- Trained in emergency response, diver rescue, and in-water incident management

Open Water Scuba

2024 (Active)

NATIONAL ASSOCIATION OF UNDERWATER INSTRUCTORS

- Certified in foundational scuba diving, navigation, and safety procedures